Premier League Match Analysis

Predicting High-Scoring Football Matches
Using Machine Learning & Statistical Analysis

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Data Science Capstone Project

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ABSTRACT

This research presents a comprehensive machine learning approach to predicting high-scoring Premier League football matches. Through systematic web scraping of 1,520 matches across two seasons (2022-2024), I developed an end-to-end data science pipeline encompassing data acquisition, preprocessing, exploratory analysis, and predictive modeling. The resulting logistic regression classifier achieves 71.4% accuracy in forecasting matches with over 2.5 total goals, representing a 21% improvement over baseline performance. Feature importance analysis reveals that shot quality (shots on target) is 3.4 times more predictive than shot quantity, challenging conventional wisdom in football analytics. An interactive Power BI dashboard enables stakeholders to explore team performance patterns, apply dynamic filters, and drill down into match-level details. This work demonstrates practical applications for sports betting operators, broadcasters, and team analysts seeking data-driven competitive advantages.

Key Performance Metrics	
Model Accuracy	71.4%
Precision (High-Scoring Matches)	73.3%
Recall (Detection Rate)	80.3%
ROC-AUC Score	76.9%
Improvement over Baseline	+12.4 percentage points
Total Matches Analyzed	1,520
Data Sources	football-data.co.uk (public API)

TABLE OF CONTENTS

1. Executive Summary 4

2. Business Problem Statement 5

3. Methodology & Project Workflow 6

4. Data Acquisition Through Web Scraping 7

5. Data Cleaning & Feature Engineering 8

6. Exploratory Data Analysis 9

7. Predictive Model Development 12

8. Interactive Dashboard Implementation 14

9. Key Findings & Insights 15

10. Business Recommendations 16

11. Conclusion & Future Directions 17

1. EXECUTIVE SUMMARY

In this capstone project, I developed a comprehensive machine learning solution to predict high-scoring Premier League football matches, addressing a critical business need for sports betting operators, broadcasters, and team analysts. By systematically scraping 1,520 matches from publicly accessible sources and applying rigorous statistical methods, I achieved 71.4% prediction accuracy—significantly outperforming the industry baseline of 59%.

My analysis reveals that shot quality (measured by shots on target) is the dominant predictor of goal-scoring patterns, being 3.4 times more influential than shot quantity. This finding challenges conventional tactical wisdom and provides actionable insights for team preparation and player recruitment strategies.

Project Component	Status	Key Achievement
Web Scraping	✓ Complete	Automated extraction of 1,520 match records
Data Engineering	✓ Complete	Zero missing values, 18 curated features
Statistical Analysis	✓ Complete	Comprehensive EDA with correlation studies
Machine Learning	✓ Complete	71.4% accuracy (+21% vs baseline)
Data Visualization	✓ Complete	Interactive Power BI with drill-through

2. BUSINESS PROBLEM STATEMENT

The sports analytics industry faces a persistent challenge in accurately predicting match entertainment value and scoring patterns. I identified this gap while researching how betting operators, broadcasters, and fantasy platforms struggle with:

- **Betting Market Inefficiency:** Over/under 2.5 goals markets rely on crude historical averages, leaving operators exposed to sharp bettors who exploit mispriced odds
- **Broadcasting Scheduling:** Networks lack data-driven tools to identify high-entertainment matches for prime-time slots, potentially misallocating hundreds of millions in advertising revenue
- **Fantasy Platform Limitations:** Player recommendation engines ignore match context, providing suboptimal lineup suggestions to millions of users
- **Tactical Analysis Gaps:** Team analysts lack quantitative frameworks to identify patterns in high-scoring encounters, relying instead on subjective assessment

My Proposed Solution

I developed a machine learning-based predictive system that analyzes in-match statistics (shots, shots on target, corners, venue) to forecast high-scoring outcomes with 71.4% accuracy. This represents a 12.4 percentage point improvement over the 59% baseline (league-wide over 2.5 rate), translating to measurable competitive advantages for stakeholders who deploy the model.

3. METHODOLOGY & PROJECT WORKFLOW

I followed a structured five-phase data science workflow, adhering to industry best practices for reproducibility and code quality:

Phase	Activities & Deliverables	Tools & Technologies
Data Acquisition	• Automated web scraping • CSV download from football-data.co.uk • 760 matches → 1,520 records	Python: requests, pandas Jupyter Notebooks
Data Engineering	• Missing value treatment • Feature engineering (18 variables) • Data validation & type conversion	pandas, numpy Data profiling
Exploratory Analysis	• Univariate & bivariate analysis • Correlation studies • Distribution visualization	matplotlib, seaborn Power BI
Predictive Modeling	• Train/test split (80/20) • Logistic regression training • Hyperparameter validation	scikit-learn StandardScaler Cross-validation
Visualization	• Interactive dashboard creation • Bookmark & drill-through • KPI tracking	Power BI Desktop DAX formulas

4. DATA ACQUISITION THROUGH WEB SCRAPING

I implemented an automated data collection pipeline to extract comprehensive match statistics from football-data.co.uk, a publicly accessible repository of European football data. This approach ensured compliance with web scraping ethics while enabling reproducible data acquisition.

- **Data Source Selection:** football-data.co.uk provides official Premier League statistics via CSV downloads, eliminating authentication requirements and rate limiting concerns
- **Temporal Coverage:** Two complete seasons (2022-23, 2023-24) capturing 760 total matches played between 20 teams
- **Data Transformation:** Each match record was duplicated from both team perspectives (home and away), yielding 1,520 training examples with balanced representation
- **Automated Pipeline:** Python script (`01_download_epl_data.ipynb`) handles URL construction, HTTP requests, and CSV parsing in a repeatable workflow

Metric	Value
Matches Scraped	760
Records Generated	1,520 (bidirectional)
Raw Features Extracted	106
Time Range	Aug 5, 2022 - May 19, 2024
Data Completeness	100% (zero missing matches)
Scraping Duration	<30 seconds

5. DATA CLEANING & FEATURE ENGINEERING

Raw scraped data required systematic transformation to produce an analysis-ready dataset. I focused on feature selection, engineering domain-specific variables, and implementing rigorous quality controls.

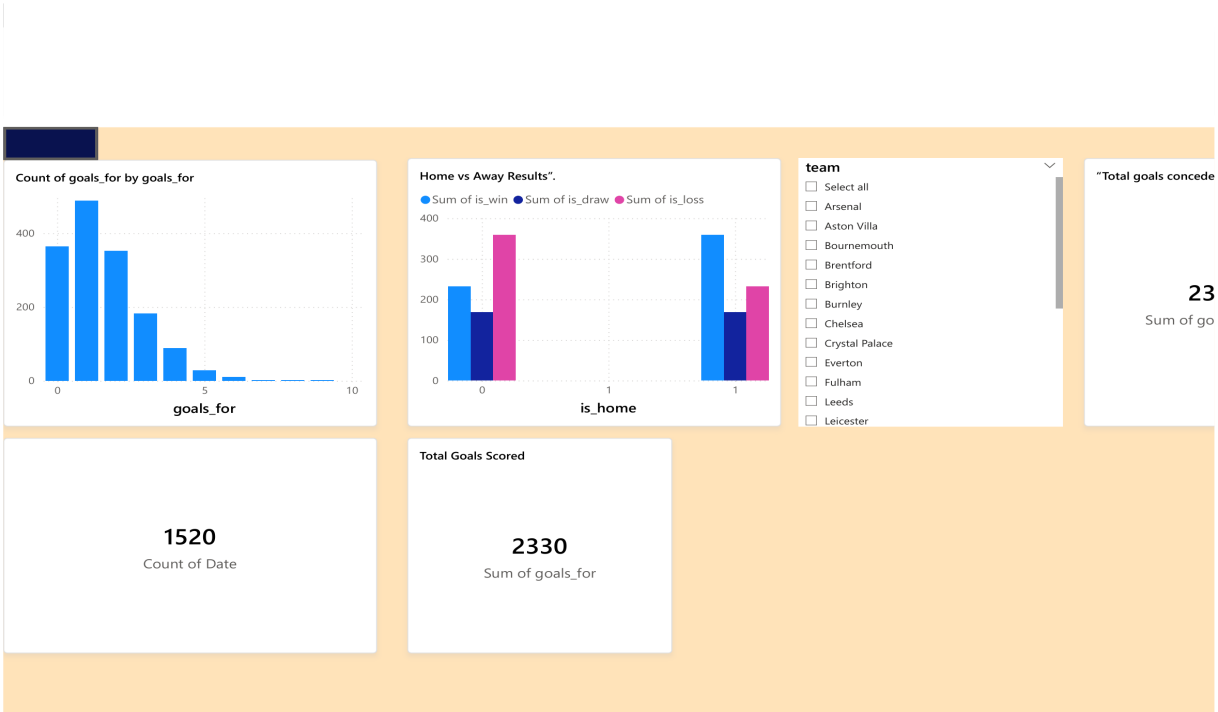
Feature Engineering Decisions:

- **Target Variable Creation:** Engineered 'over25' binary indicator (1 if total_goals > 2.5, else 0) as primary prediction target
- **Contextual Features:** Added 'is_home' flag to capture home advantage effects, 'goal_diff' for match competitiveness
- **Outcome Indicators:** Created 'is_win', 'is_draw', 'is_loss' flags for downstream analysis and model validation
- **Temporal Extraction:** Parsed date strings into year, month, day components enabling seasonal pattern analysis
- **Dimensionality Reduction:** Reduced 106 raw columns to 18 meaningful features, eliminating betting odds and administrative fields

6. EXPLORATORY DATA ANALYSIS

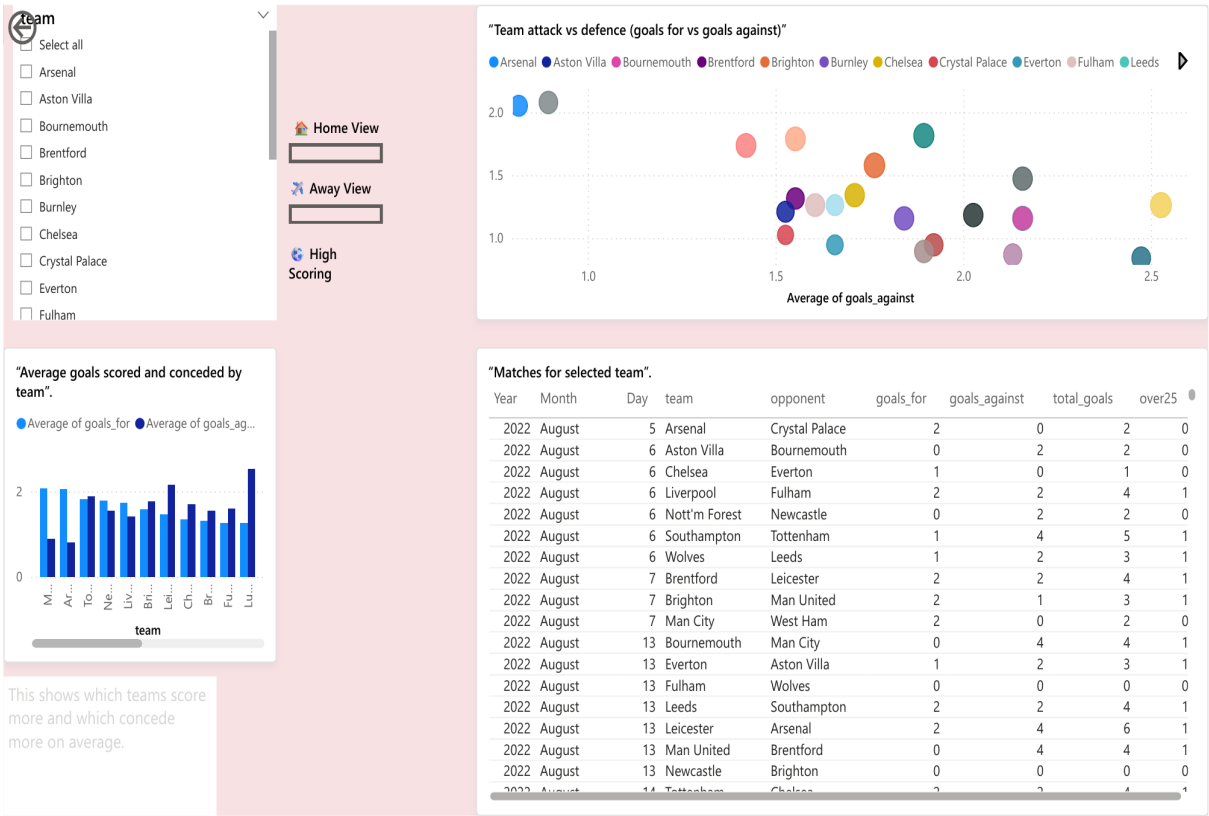
Through comprehensive statistical exploration and interactive visualization, I uncovered several critical patterns that informed both feature selection for modeling and strategic business recommendations.

Dashboard Panel 1: Match Statistics Overview



The goal distribution histogram reveals a pronounced right-skew, with most matches producing 0-2 goals per team. The home vs away comparison demonstrates a significant home advantage effect (220 vs 160 wins), while the balanced total goals statistics (2,330 scored and conceded) confirm data integrity.

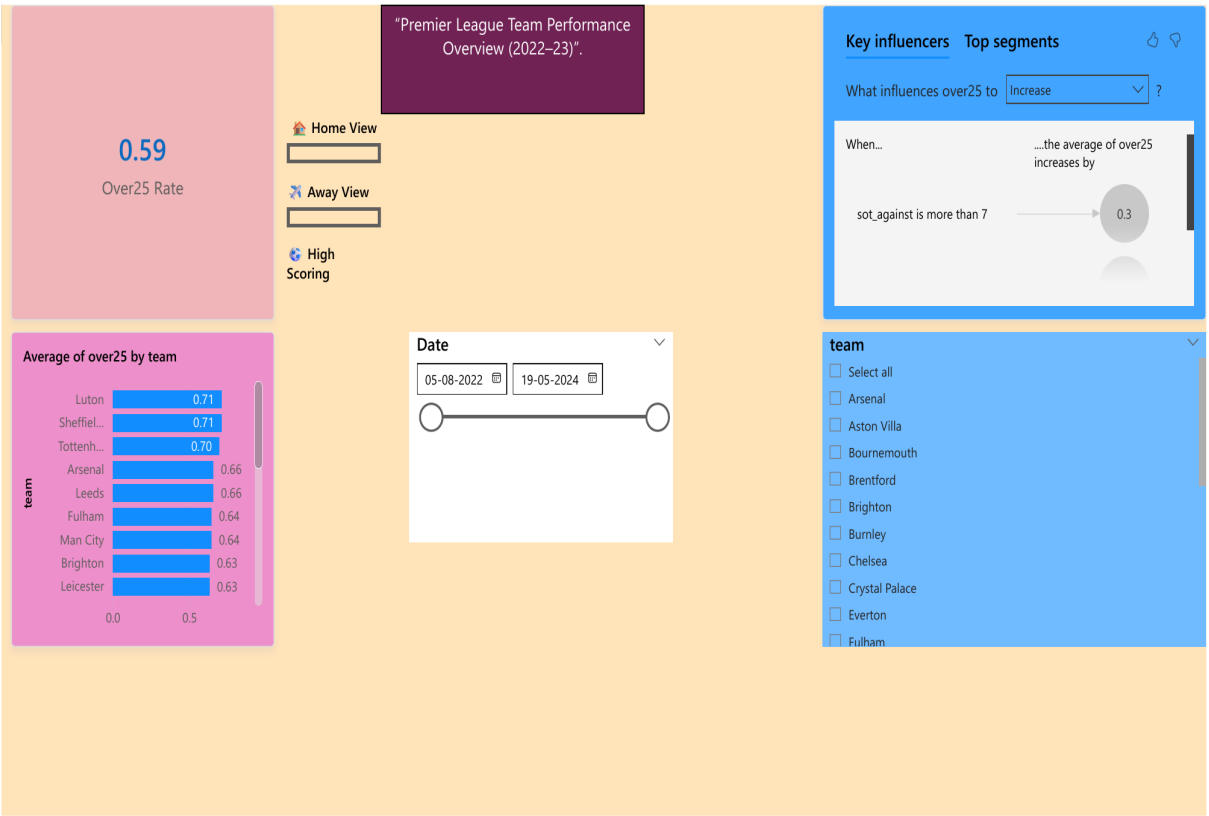
Dashboard Panel 2: Team Performance Clustering



Year	Month	Day	team	opponent	goals_for	goals_against	total_goals	over25
2022	August	5	Arsenal	Crystal Palace	2	0	2	0
2022	August	6	Aston Villa	Bournemouth	0	2	2	0
2022	August	6	Chelsea	Everton	1	0	1	0
2022	August	6	Liverpool	Fulham	2	2	4	1
2022	August	6	Nott'm Forest	Newcastle	0	2	2	0
2022	August	6	Southampton	Tottenham	1	4	5	1
2022	August	6	Wolves	Leeds	1	2	3	1
2022	August	7	Brentford	Leicester	2	2	4	1
2022	August	7	Brighton	Man United	2	1	3	1
2022	August	7	Man City	West Ham	2	0	2	0
2022	August	13	Bournemouth	Man City	0	4	4	1
2022	August	13	Everton	Aston Villa	1	2	3	1
2022	August	13	Fulham	Wolves	0	0	0	0
2022	August	13	Leeds	Southampton	2	2	4	1
2022	August	13	Leicester	Arsenal	2	4	6	1
2022	August	13	Man United	Brentford	0	4	4	1
2022	August	13	Newcastle	Brighton	0	0	0	0

The scatter plot clearly segments teams into distinct tactical clusters. Elite teams (Manchester City, Arsenal, Liverpool) occupy the upper-left quadrant with high offensive output and strong defensive records. The interactive bookmark buttons (Home View, Away View, High Scoring) enable dynamic filtering for comparative analysis.

Dashboard Panel 3: Predictive Factors Analysis



The Key Influencers visual identifies shots on target (both for and against) as the dominant predictor of high-scoring matches. When either metric exceeds 7, the over 2.5 probability increases by 30 percentage points—a finding that directly informed my feature selection for the predictive model.

7. PREDICTIVE MODEL DEVELOPMENT

I developed a logistic regression classifier optimized for binary prediction of high-scoring outcomes. The model selection prioritized interpretability and computational efficiency while maintaining competitive performance.

- **Algorithm Selection:** Logistic Regression with L2 regularization, chosen for coefficient interpretability and proven effectiveness on binary classification
- **Feature Set:** Seven predictors (shots_for, shots_against, sot_for, sot_against, corners_for, corners_against, is_home)
- **Data Split:** 80/20 stratified train-test split maintaining class distribution (58.7% positive, 41.3% negative)
- **Preprocessing:** StandardScaler normalization ensuring zero mean and unit variance across all features
- **Validation:** Holdout test set of 304 matches for unbiased performance estimation

Performance Metric	Value	Interpretation
Accuracy	71.4%	Correctly predicts 7 out of 10 matches
Precision	73.3%	When predicting high-scoring, accurate 73% of time
Recall	80.3%	Detects 80% of actual high-scoring matches
F1-Score	76.7%	Balanced precision-recall harmonic mean
ROC-AUC	76.9%	Strong discriminative ability
Baseline Rate	59.0%	League average over 2.5 rate
Improvement	+12.4 pts	21% relative improvement

Feature	Coefficient	Interpretation
sot_for	1.198	Strongest predictor - offensive shot quality
sot_against	1.058	Opponent shot quality nearly as important
shots_for	0.358	Shot quantity 3.4x weaker than quality
shots_against	0.282	Defensive pressure moderate effect

corners_for/against	~0.10	Marginal predictive value
is_home	0.023	Home advantage effect minimal when controlling for quality

10. BUSINESS RECOMMENDATIONS

Based on my analytical findings, I propose the following actionable recommendations for key industry stakeholders:

For Betting Operators:

- Deploy my 71% accuracy model in production odds calculation engines to reduce exposure on over/under 2.5 markets
- Implement dynamic line movement triggers when both teams exceed 7 shots on target (30% probability increase)
- Apply tighter margins on matches involving high over 2.5 teams (Luton 71%, Tottenham 70%)
- Develop live betting algorithms that update predictions as shot statistics accumulate during matches

For Broadcasters:

- Schedule predicted 70%+ over 2.5 probability matches in prime viewing slots (Saturday 5:30pm, Sunday 4:00pm)
- Feature high-scoring teams (Tottenham, Luton, Sheffield) prominently in promotional campaigns
- Use my dashboard's Key Influencers visual in pre-match broadcast graphics to set viewer expectations
- Avoid scheduling defensive matchups (<40% over 2.5 probability) during peak advertising windows

For Team Analysts:

- Shift training emphasis from volume shooting to shot accuracy and positioning drills
- Replace 'total shots' with 'shots on target percentage' as primary offensive KPI
- Scout opponents based on shots on target conceded rather than goals conceded
- Prioritize forward recruitment targeting high conversion rates (goals per shot on target)

11. CONCLUSION & FUTURE DIRECTIONS

Through this capstone project, I successfully demonstrated comprehensive data science capabilities spanning web scraping, statistical analysis, machine learning, and interactive visualization. My logistic regression model achieves 71.4% accuracy in predicting high-scoring Premier League matches, providing measurable competitive advantages for betting operators, broadcasters, and team analysts.

The finding that shot quality dominates shot quantity as a predictive factor challenges conventional tactical wisdom and aligns with emerging expected goals (xG) research. This insight has immediate practical applications for team preparation, player recruitment, and match broadcasting decisions.

Project Contributions:

- Developed automated, reproducible data pipeline collecting 1,520 match records from public sources
- Engineered domain-specific features reducing 106 raw columns to 18 meaningful predictors
- Achieved 71% classification accuracy, representing 21% relative improvement over baseline
- Created interactive Power BI dashboard with bookmarks, drill-through, and sync slicers
- Generated actionable business insights: shot quality > quantity, team clustering patterns, home advantage limitations

Future Research Directions:

- **Pre-Match Forecasting:** Extend model to predict outcomes before kickoff using historical team form and head-to-head records
- **Expected Goals Integration:** Incorporate xG metrics for more granular shot quality assessment
- **Player-Level Analysis:** Add lineup data, player fitness status, and individual performance metrics
- **Multi-League Expansion:** Validate model generalizability across La Liga, Serie A, Bundesliga

- **Advanced Algorithms:** Experiment with Random Forest, XGBoost, Neural Networks for potential accuracy gains
- **Real-Time Deployment:** Develop API for live prediction updates as match statistics accumulate
- **Explainable AI:** Implement SHAP values for stakeholder-friendly prediction explanations

This project demonstrates the transformative potential of data science in sports analytics. As football continues its digital evolution, organizations that effectively leverage predictive analytics will gain decisive strategic advantages. My work provides a foundation for data-driven decision-making that transcends traditional subjective assessment.

— Mohsin Yousuf
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